



# Identity

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# Identity and Early Maladaptive Schemas: A Network-Analytical Approach

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## ABSTRACT

“Who am I?” is a question all emerging adults face. Research shows that, while achieving identity synthesis is crucial for psychosocial well-being, many youngsters experience identity confusion, making them vulnerable to psychological distress. There is also growing interest in pathological identity problems, where individuals feel a complete “lack of identity.” Understanding factors linked to identity issues is key to addressing related psychopathologies. This study, based on schema theory, investigated how identity synthesis, normative identity confusion and pathological identity confusion are related to Young’s 18 early maladaptive schemas (EMS) in a sample of 711 emerging adults. Network analysis revealed that normative identity confusion was uniquely related to emotional inhibition and insufficient self-control, while pathological identity confusion was related to negativity, social isolation, emotional deprivation, and shame/defectiveness. Identity synthesis was not linked to any EMS. Results suggest EMS in the domain of disconnection and rejection are most consistently associated with identity problems, particularly pathological ones.

## KEYWORDS

Identity; early maladaptive schemas; emerging adulthood; adolescence; network analysis

## Introduction

### *Identity development*

The construct of identity has gained significant attention in psychology. Erik Erikson (1950), who is credited with establishing the groundwork for modern identity theory (Schwartz, 2001), conceptualized it as a developmental task that starts in adolescence but maintains relevance throughout the lifespan. Youngsters need to tackle questions such as “Who am I?”, “What are my values?” and “What are my goals in life?”, as they increasingly become autonomous from their parents (Beyers et al., 2024). For Erikson, identity mainly represents a subjective feeling of sameness and continuity across time and across contexts, situated on a single bipolar dimension, ranging from identity synthesis (i.e., a coherent view of-self) to identity confusion (i.e., a fragmented view-of-self). While confusion is characterized by exploratory behavior, feelings of inner emptiness and failing to come to identity commitments, synthesis is characterized by making solid and authentic identity commitments. Many researchers have built upon Erikson’s work and refined his theory. Consequently, models of identity focus on statuses (Marcia, 1966), styles (Berzonsky, 1990) or processes (Crocetti et al., 2008; Luyckx et al., 2006, 2008), yet all models carry in them the notion of the need to develop a solid and coherent sense-of-self, while acknowledging that everyone can experience periods of identity confusion.

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Central to Eriksons theory is that constructing a strong sense of self is essential to psychosocial well-being. Research has indeed shown that when people have a clearer view of their identity, they tend to report more life satisfaction and psychological well-being (Hatano et al., 2022; Schwartz et al., 2011). On the other hand, identity confusion in individuals has been related to higher rates of psychopathology (e.g., Crocetti et al., 2009, Ghandi et al., 2019; Verschueren et al., 2017).

Historically, research on identity has predominantly approached the concept from a developmental perspective, with clinical research largely overlooking the concept. Consequently, the conceptualization and measurement of identity has primarily been approached from a “normative” point of view (Schwartz, 2001, 2014). However, individuals with psychopathology frequently diverge from normative developmental trajectories. In an effort to integrate developmental and clinical perspectives, Kaufman et al. (2015) highlighted the necessity to differentiate between normative turbulence in identity development (i.e., every adolescent goes through a period of identity confusion) and pathological manifestations of identity confusion. Pathological identity confusion may present itself as feeling empty, broken or fragmented, holding conflicting values simultaneously and easily letting go of what was previously perceived as strong commitments. Such severe identity problems are a core characteristic of personality disorders, in particular borderline personality disorder (BPD; Esmaeilian et al., 2019, 2022; Kaufman & Meddaoui, 2021; Westen et al., 2011). Indeed, complete lack of a sense of identity, specifically “sharp discontinuities in self-concept characterized by shifting goals, values and vocational aspirations, sudden changes in opinions and plans about career, sexual identity, values and types of friends” has been taken on in the DSM-5 as a core criterium for all personality disorders (American Psychiatric Association [APA], 2013). Research corroborates this idea, as one study asked participants to repeatedly list personality traits within a three-hour period. Despite this short time-frame, participants with BPD compared to control participants were more likely to change their self-descriptions across the trials (Beeney et al., 2016), attesting to the lack of sameness and continuity of identity over time and contexts, as Erikson describes as evident in identity synthesis. Taken together, it has been argued that identity needs to be viewed as a bipolar dimension ranging from synthesis to confusion, and that this confusion can be normative as well as pathological in nature. Making the distinction between normative and pathological identity confusion is important. First of all, because it allows for more nuanced descriptions of the heterogeneous experiences of identity, and also because different factors may be underlying these experiences.

Despite the importance of identity for health (Hatano et al., 2022; Schwartz et al., 2011), surprisingly little is known about what factors underly identity development. There is some research which points to the role of attachment or early childhood adversities (Gandhi et al., 2016, 2019; Hayward et al., 2020; Vartanian et al., 2018). However, this research falls short to take into account the heterogeneity in identity experiences (synthesis, normative confusion, pathological confusion) and, as such, the complexity of factors underlying these different identity experiences. The present study used schema theory (Young et al., 2004) to take a closer look at this issue and argued that Early Maladaptive Schemas (EMS) may underlie synthesized or (pathologically) confused identity development.

### ***Early maladaptive schemas as underlying mechanisms of identity development?***

Youngs defines schemas as “a broad, pervasive theme or pattern, comprised of memories, emotions, cognitions, and bodily sensations, regarding oneself and one’s relationships with others, developed during childhood or adolescence, elaborated throughout one’s lifetime and dysfunctional to a significant degree” (Young et al., 2004, p. 7). Central to Youngs notion of schemas is his developmental approach to the concept. He states that every person has core emotional needs. Maladaptive schemas, he argues, are the result of a person’s attempt to make sense of repeated need frustration because of negative social experiences while growing up (hence they are referred to as “Early” Maladaptive Schemas). Early in life, these negative experiences are mainly experienced with immediate family members, while later on the

influences of peers and society grow. As such, EMS are inherently psychosocial in nature since they develop in relation to others and, once developed, constitute a framework for thoughts, feelings and action for interpersonal interactions.

Currently, 18 EMS are identified and included in a measurement tool, the Young Schema Questionnaire (YSQ; Young & Brown, 2005). In his original theory (Young et al., 2004), these 18 schemas were organized into five themes called “domains,” with each domain referring to a frustrated emotional need. With empirical testing of this model and subsequent theoretical refinement, there is growing consensus among researchers that four domains best represent the schemas higher-order structure, these being *Disconnection & Rejection*, *Impaired Autonomy & Performance*, *Excessive Responsibility & Standards* and *Impaired Limits* (Bach et al., 2017; Thimm, 2022; Young & Arntz, 2014). See Table 1 for an overview and description of these four domains.

Referencing Eriksons work, Young et al. (2004) posit that EMS and resolution of developmental tasks are closely related. As EMS make people vulnerable for psychopathology (Dozois et al., 2013; Maher et al., 2022; Tariq, et al., 2021a; 2021b) and interpersonal problems (Barazandeh et al., 2016; Esmaeilian et al., 2019; Zeigler-Hill et al., 2011), they subsequently hamper positive resolution of developmental tasks later on in life. For the development of a synthesized identity, this might mean that EMS developed in childhood and early adolescence may hamper the development of a coherent self-concept and underlie (pathological) identity confusion.

Indeed, the sparse literature that is available on the identity-EMS relationship demonstrates that identity is related to schemas situated in multiple schema domains. Thimm (2009), in a sample of 145 psychiatric outpatients, linked identity synthesis with lower scores on EMS in all schema domains. Esmaeilian et al. (2019), investigating EMS in relation to BPD characteristics in a community sample, found that identity problems were uniquely related to the schemas of abandonment (disconnection & rejection domain), dependence/incompetence, vulnerability to harm (impaired autonomy & competence domain) and insufficient self-control (impaired limits domain). A follow-up study (Esmaeilian et al., 2022) again showed unique associations with the abandonment schema. As such, both theory and preliminary research suggest that identity development is related to EMS, however it remains unclear how specific EMS relate to different forms of identity functioning, ranging from synthesis to pathological confusion.

The goal of the present study was to explore the specific relationships between EMS and identity functioning. To capture the heterogeneity of identity experiences, relationships with both normative identity confusion and pathological identity confusion were investigated alongside identity synthesis. In doing so, this study hoped to provide a more nuanced and in-depth analysis of the factors underlying synthesis and confusion.

As this is an exploratory study and given the potentially complex relationships between EMS and different forms of identity functioning (Esmaeilian et al., 2019; Thimm, 2009), a method is needed that can capture such complexity. Traditional variable-centered approaches typically focus on overall effects or latent structures, but they are less suited to disentangling the unique associations among multiple interrelated constructs. Network analysis, by contrast, is a data-driven method that models multivariate dependencies directly, offering a comprehensive view of how constructs are uniquely connected, especially when complex associations can be expected (Borsboom et al., 2021). In particular, we will model unique associations between identity functioning (synthesis, normative confusion and pathological confusion) and EMS using a community sample of late adolescents and emerging adults. In a network model, variables are represented as “nodes,” while the unique relationships between them are illustrated by “edges” that link these nodes together. More influential nodes occupy a central position within the network, and the overall structure of the network can provide valuable insights into the complex interactions between different constructs. This method has gained popularity in psychological research for examining intricate patterns of connectivity between psychological variables (e.g., Hoorelbeke et al., 2016, 2019). Regarding identity, Esmaeilian et al. (2019, 2022) conducted two network studies exploring the distinct associations between borderline personality features – including a node representing identity problems – and EMS. However, no research to date

**Table 1.** Early maladaptive schemas, their respective domains and internal consistency of the scale in the present study.

| Domain                               | Description                                                                                                                                                                                                                                                                                                                                                                          | Schema                    | Example item                                                                                                         | Cronbach's Alpha |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------|------------------|
| Disconnection & Rejection            | This domain involves core beliefs that one's basic emotional needs for security, love, acceptance, and belonging will not be met. People with schemas in this domain often fear rejection, abandonment, humiliation, or feel emotionally deprived and fundamentally different from others.                                                                                           | Emotional deprivation     | "I haven't had someone to nurture me, share him/herself with me, or care deeply about everything that happens to me" | $\alpha = .79$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Social Isolation          | "I don't fit in"                                                                                                     | $\alpha = .89$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Emotional Inhibition      | "I find it hard to be free-spirited and spontaneous around other people"                                             | $\alpha = .82$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Defectiveness/Shame       | "No one I desire would want to stay close to me if he or she knew the real me"                                       | $\alpha = .87$   |
| Impaired Autonomy & Performance      | This domain involves core beliefs that one is incapable of managing everyday responsibilities, making decisions, or achieving success without significant help. People with schemas in this domain often struggle with dependence, feelings of failure, vulnerability, or a weak sense of identity and direction.                                                                    | Mistrust/Abuse            | "I feel that people will take advantage of me"                                                                       | $\alpha = .86$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Negativity/Pessimism      | "Even when things seem to be going well, I feel that it is only temporary"                                           | $\alpha = .83$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Dependence/Incompetence   | "I do not feel capable of getting by on my own in everyday life"                                                     | $\alpha = .77$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Failure                   | "I'm not as talented as most people are at their work"                                                               | $\alpha = .88$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Subjugation               | "I think that if I do what I want, I'm only asking for trouble"                                                      | $\alpha = .76$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Abandonment/Instability   | "I need other people so much that I worry about losing them"                                                         | $\alpha = .86$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Enmeshment                | "I often feel I do not have a separate identity from my parent(s) or partner"                                        | $\alpha = .72$   |
| Excessive Responsibility & Standards | This domain involves core beliefs that one's worth depends on meeting high internal standards or putting others' needs above one's own. People with schemas in this domain often feel driven by perfectionism, guilt, or pressure to be in control or overly self-sacrificing, often at the cost of their own well-being.                                                            | Vulnerability to harm     | "I feel that a disaster (natural, criminal, financial, or medical) could strike at any moment"                       | $\alpha = .78$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Self-Sacrifice            | "I'm so busy doing things for the people that I care about that I have little time for myself"                       | $\alpha = .81$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Unrelenting Standards     | "I must meet all my responsibilities"                                                                                | $\alpha = .75$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Self-Punitiveness         | "I'm a bad person who deserves to be punished."                                                                      | $\alpha = .79$   |
| Impaired Limits                      | This domain involves core beliefs related to difficulty respecting boundaries, exercising self-discipline, and setting realistic goals or limits for oneself. People with schemas in this domain often struggle with impulsivity, entitlement, or difficulty tolerating frustration, which can lead to problems with responsibility, goal achievement, or respecting others' rights. | Entitlement/Grandiosity   | "I feel that I shouldn't have to follow the normal rules or conventions that other people do"                        | $\alpha = .57$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Approval Seeking          | "Lots of praise and compliments make me feel like a worthwhile person"                                               | $\alpha = .71$   |
|                                      |                                                                                                                                                                                                                                                                                                                                                                                      | Insufficient Self-Control | "I have rarely been able to stick to my resolutions"                                                                 | $\alpha = .78$   |

Note. Domains and their content are based on Bach et al. (2017).

has directly examined the unique connections between different aspects of identity functioning and EMS.

Given that this is the first study to investigate the unique relations between different forms of identity functioning and EMS, our hypotheses were largely exploratory. Broadly speaking, we expected EMS to be negatively associated with identity synthesis and positively associated with identity confusion (normative and pathological). Moreover, we had no clear hypotheses on which unique EMS-identity associations we would find, but based on previous research on borderline personality features, we expected that the pathological identity confusion would predominantly show unique associations with EMS within the domain of disconnection/rejection (Esmailian et al., 2019, 2022).

## Method

### *Participants and procedure*

Throughout the summer of 2023 a community sample of 711 late adolescents and emerging adults was recruited. The sample included senior year secondary education students (47.1%; recruited through high schools), higher education students (44.9%; recruited through the institutions' communication channels) and non-studying/working emerging adults (8%; recruited online). Both high schools and higher education institutions involved in this study were publicly funded and urban-situated. The procedure was the same for all participants and consisted of filling out an online test battery in Dutch consisting of questionnaires relating EMS and identity, as well as demographics, which took approximately 40 min. All participants gave informed consent before filling out an online questionnaire. Parents of participants who were still minors at the time of participation (i.e., those in high school) were informed and could decide for their child to opt out of the study. Gift certificates were raffled to incentivize participation. Instructed response items (e.g., "choose response option four") were added to assess whether participants filled out the questionnaires in a focused manner. In total, 1258 people responded to the questionnaire. Respondents were excluded if they did not completely fill out the YSQ or did not correctly respond to two out of three instructed items. Finally, a sample of 711 youngsters remained, consisting of 68.9% females (30.1% male; 1% "other") with an average age of 19.53 years ( $SD = 1.88$ ; range = 16.38–24.84 years). The use of a community sample allows us to capture greater heterogeneity in identity structures, and emerging adults are chosen as the population of interest because of the distinct relevance of identity as a developmental task in this age group (Arnett, 2000). Related to this, personality disorders often find their onset in adolescence/emerging adulthood (e.g., Chanen & Thompson, 2019; Videler et al., 2019).

This procedure was approved by the Ethical Commission of the authors' institution (Ghent University) [Approval Number: 2022–104] and the study was performed in accordance with the ethical standards as laid down in the 1964 Declaration of Helsinki and its later amendments or comparable ethical standards.

## Measures

### *Identity*

In this study, identity synthesis, normative identity confusion and pathological identity confusion were measured. For the first two, the identity subscale in the Erikson Psychosocial Stage Inventory (EPSI; Rosenthal et al., 1981) was used, which contains twelve items, six to assess *identity synthesis* and six to assess *identity confusion*. Respective example items include "I know what kind of person I want to be" and "I don't really know who I am." Statements were rated using a 5-point Likert scale, ranging from 1 (*strongly disagree*) to 5 (*strongly agree*). The (Dutch version of the) EPSI has been found to have good psychometric qualities (Bogaerts et al., 2019; Schwartz et al., 2011; Verschueren et al., 2018). In

this study, internal consistency using Cronbach's Alpha was .70 (identity synthesis) and .77 (identity confusion). To measure pathological identity confusion, the *lack of identity* subscale of the Self-Concept and Identity Measure (SCIM; Kaufman et al., 2015) was included to tap into pathological identity problems. Example items include "I feel lost when I think about who I am" and "I am broken" and were answered on the same Likert scale. Good psychometric properties have been found for the Dutch version of the questionnaire (Verschuere et al., 2018). Cronbach's Alpha for this measure was .91 in the current study.

### Early maladaptive schemas

The Young Schema Questionnaire – Version 3 Short Form (YSQ-S3; Young & Brown, 2005) aims to measure maladaptive schemas. The questionnaire consists of 18 subscales made up of five items each. A 6-point Likert Scale ranging from 1 (*completely untrue of me*) to 6 (*describes me perfectly*) is used. Previous studies have shown that the YSQ-S3 has good internal consistency and validity (e.g., Bach et al., 2015; Bosmans et al., 2010; Rijkeboer et al., 2005). Example items and internal consistencies for each subscale are presented in Table 1.

### Plan of analyses

Data analysis was performed in R Version 4.3.1 (see Supporting Information for R package version information). We employed a nonparanormal transformation using the "huge" package to improve normality (Epskamp & Fried, 2018; also see; Zhao et al., 2015). Subsequently, we utilized the "qgraph" package (Epskamp et al., 2012) to estimate a Gaussian Graphical Model (GGM; see Epskamp & Fried, 2018). To reduce spurious edges, we employed the graphical least absolute shrinkage and selection operator (GLASSO; Friedman et al., 2014) with Extended Bayesian Information Criterion model selection (EBIC;  $\gamma = 0.5$ ). We implemented additional thresholding to obtain a sparse regularized partial correlation network.

The obtained GGM was visualized using the "qgraph" package, employing a variant of the Fruchterman – Reingold algorithm (Fruchterman & Reingold, 1991). This algorithm aims to arrange nodes in the network based on their level of connectivity: nodes with stronger connections occupy more central positions, while those with weaker connections are placed in the periphery of the graph. Edges represent the specific associations between two nodes, with their thickness and color indicating the strength and valence of the association (blue and full lines for positive, red and dashed lines for negative associations). A pie chart around each node visualizes the percentage of variance ( $R^2$ ) explained by neighboring nodes. To enhance the comprehension of the relationship between identity functioning and EMSs, we created flow diagrams for identity synthesis, identity confusion, and lack of identity using the derived GGM network structure. These diagrams are not new network models but illustrate the exclusive connections between a particular form of identity functioning and the other nodes within the partial correlation network model.

Bootstrapping procedures (Epskamp et al., 2018) were applied to assess the precision of edge weights using the "bootnet" package (Epskamp & Fried, 2020; see supplemental material). We explored the sampling variability in edge weights, providing 95% confidence intervals for all edges within the model and highlighted significant disparities between edge weights.

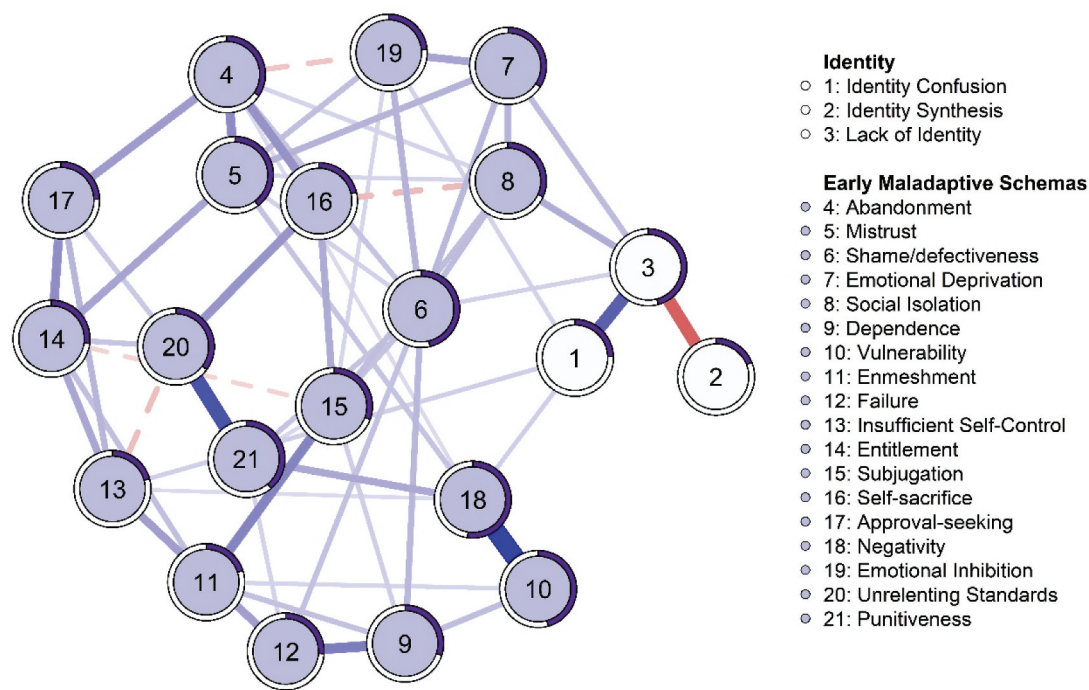
## Results

Table 2 presents descriptive statistics for the variables of interest (see supplemental table S3 for the descriptive statistics for gender, age groups and education level); Figure 1 contains the obtained GGM depicting unique associations between the three forms of identity functioning (Nodes 1–3) and EMS. The GGM contained 56 edges, most of which depict unique associations between the different EMS (85%). The strongest edge (.57) appeared between the EMS of vulnerability and negativity, being significantly stronger than all other edges included in the model (Supplemental Figure 2). Only six



**Table 2.** Descriptives and explained variance of each variable/node.

| Variable/Node             | Mean | Standard Deviation | R <sup>2</sup> |
|---------------------------|------|--------------------|----------------|
| Identity Confusion        | 2.71 | .74                | .24            |
| Identity Synthesis        | 3.54 | .77                | .19            |
| Lack of Identity          | 2.87 | 1.39               | .46            |
| Emotional deprivation     | 2.69 | 1.07               | .34            |
| Social Isolation          | 2.61 | 1.15               | .31            |
| Emotional Inhibition      | 2.75 | 1.09               | .24            |
| Defectiveness/Shame       | 2.24 | 1.08               | .46            |
| Mistrust/Abuse            | 2.61 | 1.09               | .40            |
| Negativity/Pessimism      | 2.74 | 1.11               | .52            |
| Dependence/Incompetence   | 2.23 | .88                | .30            |
| Failure                   | 2.51 | 1.03               | .26            |
| Subjugation               | 2.43 | .92                | .31            |
| Abandonment/Instability   | 2.92 | 1.19               | .34            |
| Emneshment                | 2.25 | .88                | .20            |
| Vulnerability to harm     | 2.34 | 1.03               | .46            |
| Self-Sacrifice            | 3.59 | 1.02               | .21            |
| Unrelenting Standards     | 3.56 | .99                | .35            |
| Self-Punitiveness         | 2.91 | .97                | .39            |
| Entitlement/Grandiosity   | 2.70 | .72                | .27            |
| Approval Seeking          | 3.23 | .88                | .25            |
| Insufficient Self-Control | 2.99 | .97                | .21            |



**Figure 1.** Network model. Note: This Gaussian Graphical Model presents the unique associations between the different nodes. Within this model, nodes are positioned so that well connected nodes take a more central position. Each edge in the network represents a (regularized) partial correlation, reflecting the association between two given nodes after controlling for all other nodes in the model. Red lines indicate negative associations, blue lines positive associations. The level of explained variance for each node in the model by all directly connected nodes is represented as a pie chart around the node (in dark blue).



unique associations (13%) were identified between identity features and EMS. Notably, identity confusion and lack of identity exhibit distinct patterns of associations with EMS, unlike identity synthesis, which is solely linked with other identity measures. Furthermore, a substantial portion of variance in identity functioning is explained by neighboring nodes, with a predictability of 24% and 46% for identity confusion and lack of identity, respectively (see Table 2). About 19% of variance in identity synthesis is accounted for by neighboring nodes, none of which include EMSs.

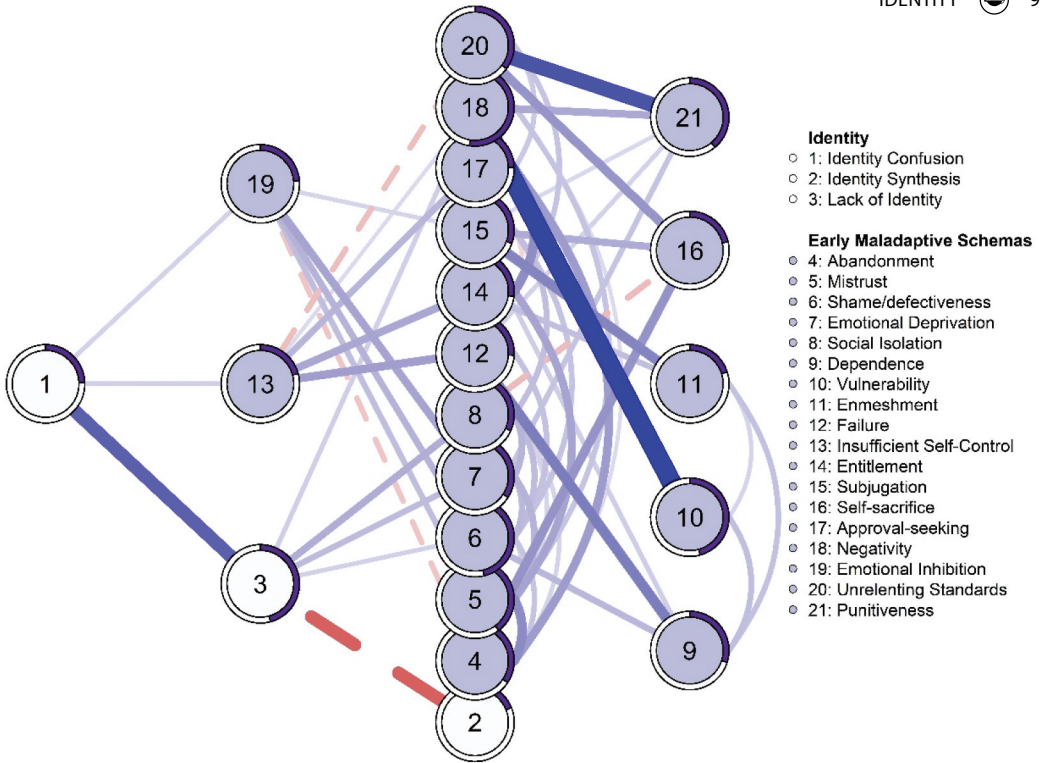
Based on the GGM, flow charts for identity synthesis, identity confusion, and lack of identity were plotted to illustrate their unique connectivity patterns with EMS. Identity confusion exhibits strong associations with lack of identity (.37) and uniquely relates to the EMS of insufficient self-control (.11) and emotional inhibition (.10; see Figure 2a). Conversely, identity synthesis is negatively associated with a lack of identity (−.36) but lacks unique associations with any EMS (see Figure 2b). Finally, lack of identity is linked with both synthesis (−.35) and confusion (.37) as well as uniquely associated with the EMS of negativity (.11), social isolation (.18), emotional deprivation (.14), and shame/defectiveness (.10; see Figure 2c). A full list of edge weights, their accuracy as well as the correlation matrix can be found in supplemental Table S1, Supplemental Figure F1 and Supplemental Table S2.

## Discussion

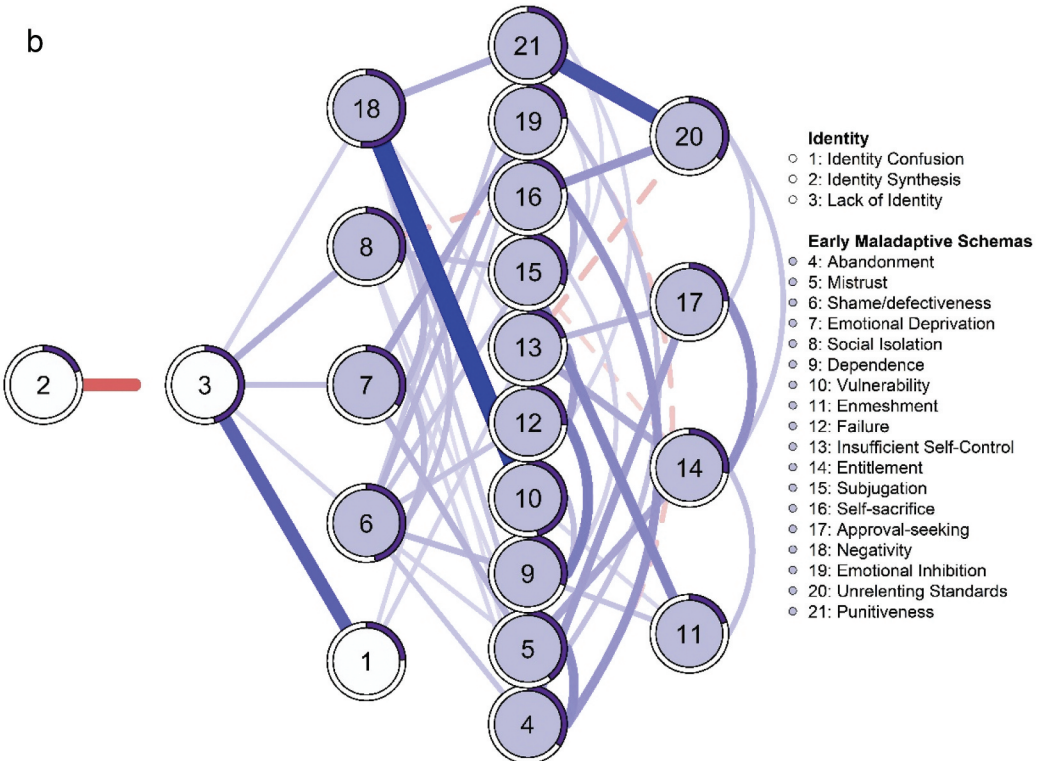
Constructing a strong sense of identity is crucial for psychosocial well-being (Schwartz et al., 2011). Identity problems have been linked to various forms of psychopathology (Schwartz et al., 2011) and are a key feature of personality disorders (Barazandeh et al., 2016; Esmailian et al., 2019; Zeigler-Hill et al., 2011). Therefore, understanding the underlying factors associated with adaptive or maladaptive identity functioning is essential. For this purpose, the current study focused on the role of EMS, as these are theorized to influence individuals' ability to navigate developmental tasks throughout life (Young et al., 2004). To assess the relationships between identity synthesis, normative identity confusion, pathological identity problems, and Young's 18 EMS, a network analysis was conducted. This method provides a data-driven but clear depiction of intricate and complex associations. To our knowledge, this is the first study to directly examine heterogeneous identity functioning in relation to EMS in emerging adults.

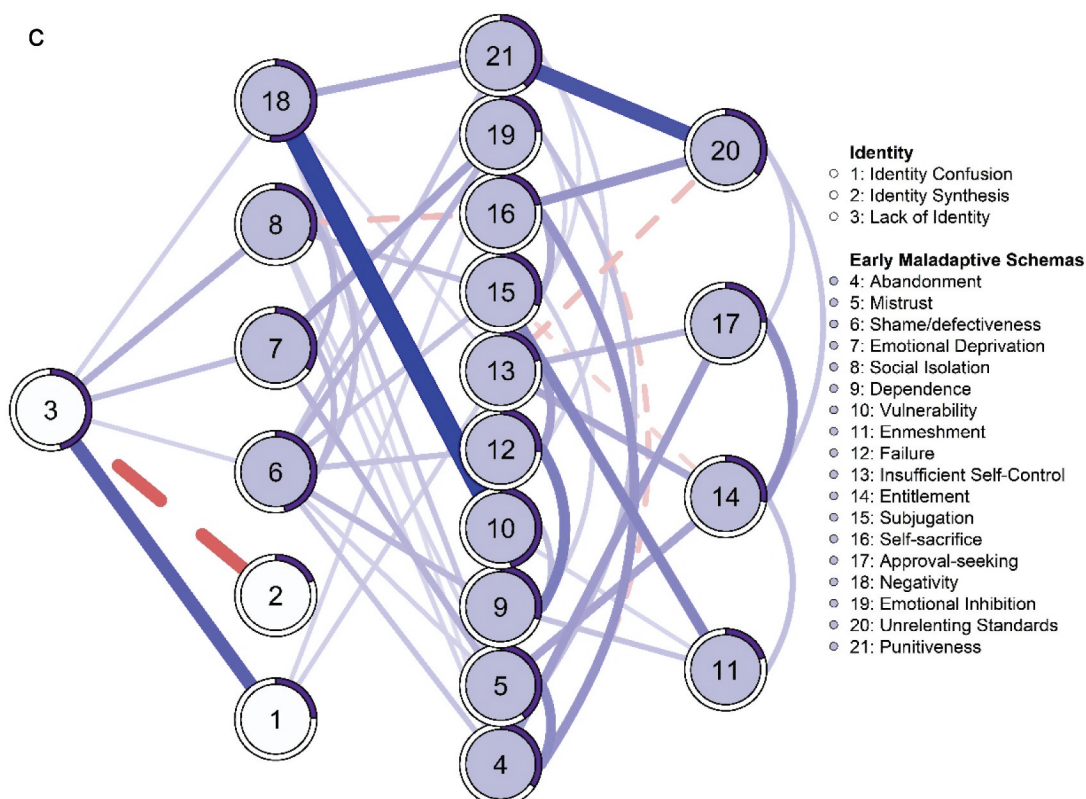
When looking at its unique associations with EMS, *pathological identity confusion* (i.e., as measured with the "lack of identity scale" of the SCIM) was related to the schemas of emotional deprivation, shame/defectiveness, social isolation, and negativity. Although the exploratory nature of this study warrants caution against making definite or causal conclusions, it is still possible to make meaningful interpretations about these associations. Emotional deprivation refers to the expectation that no one will provide attention and warmth when nurturing is needed, share and understand feelings, offer a sympathetic ear when empathy is wanted, or give guidance and strength when protection is warranted. Consequently, individuals with this schema have difficulty recognizing, trusting, or addressing their own emotional needs. Identity development, however, requires the ability to explore personal values, goals, and directions in life. Being out of touch with one's own needs may therefore undermine this process. In other words, when young people are expected to take increasing responsibility for autonomous life choices (Arnett, 2000), those who cannot clearly sense or articulate what they want or feel may face greater challenges in constructing a coherent identity. As a result, they may become vulnerable to pathological identity confusion. Shame/defectiveness refers to the belief that some part of oneself is inherently unlikable, bad, or shameful, which makes one unworthy of love or acceptance. Because identity development requires openly exploring who one is and what one values, individuals who fear that their flaws will be exposed may avoid exploration or hesitate to commit to goals and relationships. Similarly, the schema of negativity involves the expectation that life will inevitably bring failure or disappointment. Such an outlook may hinder the curiosity, optimism, and openness needed for exploration, thereby contributing to identity problems. Finally, the schema of social isolation entails feeling fundamentally different from others and cut off from community. Since identity formation is closely tied to developing a sense of belonging and locating oneself within social

a



b





**Figure 2.** Hierarchical flow charts. Note: These hierarchical flow charts are based on the network structure presented in Figure 1 and depict direct and indirect links between the core identity features and other nodes in the model. Figure 2a presents the shortest paths between Identity Confusion and related Identity features and Early Maladaptive Schemas; Figure 2b and 2c present similar paths starting from Identity Synthesis or Lack of Identity respectively. Edges between nodes reflect regularized partial correlations; the pie chart surrounding each node represents the level of variance of that node explained by all directly connected nodes.

groups (Erikson, 1950), this disconnection may make one particularly vulnerable to pathological identity confusion.

More *normative* experiences of *identity confusion* were related to the EMS of insufficient self-control and emotional inhibition. People with the schema of insufficient self-control experience problems tolerating frustrations, working toward long-term goals, and restraining the expression of impulses and excessive feelings. They avoid inconvenience as well as responsibilities, often at the expense of personal fulfillment, commitment, and integrity. Given this, it is not surprising that individuals scoring higher on this schema may also change their opinion about themselves more often, feel mixed up about who they are, or fail to feel involved in anything larger than themselves (i.e., what Erikson describes as identity confusion). Esmaeilian et al. (2019), in a sample of adults, also found a relation between identity problems and the insufficient self-control EMS. To make solid and fulfilling commitments, a person also needs to express their needs, ideas, and passions without fear of criticism or punishment. People with the schema of emotional inhibition refrain from such expression. As a result, they may become stuck in endless cycles of identity exploration without ever reaching a synthesized and coherent self-concept. Moreover, one's social identity is strongly dependent on the recognition and affirmation of identity choices by others (Erikson, 1950). Thus, an inability to express certain parts of one's identity may undermine opportunities to feel a sense of belonging.

Interestingly, *identity synthesis* showed associations only with other measures of identity functioning and did not exhibit any unique relationships with EMS. This suggests that the experience of

identity synthesis may not depend on schema scores. Although we should be cautious in drawing causal conclusions, these findings may indicate that changes in EMS are reflected primarily in maladaptive identity processes rather than adaptive ones. However, the lack of associations may also be due to the exclusive focus on maladaptive schemas. In recent years, there has been a growing interest in positive schemas (Louis et al., 2018, 2023). Positive schemas are believed to arise from emotional need satisfaction (Young et al., 2004) rather than frustration, and they contribute to psychological well-being (Keyfitz et al., 2013; Louis et al., 2018). It is conceivable that a relationship between positive schemas and a synthesized identity structure could be found. Another interesting finding was that identity synthesis related only to pathological, but not normative, confusion. Theoretically, this may reflect that normative confusion represents exploration and uncertainty that can co-exist with synthesis as part of normative development, whereas pathological confusion reflects a qualitatively different process of fragmentation and incoherence, for which synthesis may serve as a protective factor. Statistically, in a network model, edges reflect conditional associations. Because pathological confusion is the closest conceptual counterpart to synthesis, the variance they share may have absorbed the smaller overlap between synthesis and normative confusion, leaving no unique edge. Moreover, a possible relationship between identity synthesis and normative confusion can also be diluted due to their somewhat lower internal consistencies compared to pathological confusion.

Looking at the EMS on the domain level and using the four-domain model proposed by Bach et al. (2017), all EMS connected with identity problems, both normative and pathological, were situated within the disconnection/rejection domain, with the exception of insufficient self-control, which falls within the impaired limits domain. This pattern suggests that difficulties in attachment and forming secure bonds may be especially relevant for youngsters struggling with identity problems. From the perspective of attachment theory, a secure relationship provides a “secure base” for exploration and a “safe haven” for comfort and protection (Ainsworth, 1979; Bowlby, 1969). These functions are typically offered by caregivers, but teachers or other significant figures can also fulfill them. Identity development requires exploring one’s values, goals, and self-concept, while having supportive relationships that provide guidance and emotional safety. Without such a secure base, youngsters may feel unable or unwilling to explore their identity, and without a secure haven, they may be hesitant to make autonomous identity commitments. Consequently, this could contribute to more severe, pathological forms of identity confusion, and to a lesser degree, normative identity struggles. These results align with previous longitudinal research showing that supportive parenting and secure attachments promote adaptive identity development, whereas alienation in these relationships is linked to identity confusion (Beyers & Goossens, 2008; Gandhi et al., 2016, 2019; Meeus, 2011). They also correspond with findings in BPD research, where identity disturbance co-occurs with fears of abandonment and the abandonment schema emerges as a consistent correlate (Esmaeilian et al., 2019, 2022; Lawrence et al., 2011; Nilsson et al., 2010). While the present study did not find a unique association between identity problems and this specific EMS, we did observe multiple associations between identity problems and EMS belonging to the broader domain of disconnection/rejection.

### ***Strengths and limitations***

This study has several strengths. To the best of our knowledge, it is the first study to investigate the identity-EMS relationship directly. It looks at this relationship in detail, considering heterogeneity in identity and using innovative techniques to analyze the complex interrelations. This study also adds to the growing literature on identity as a clinical concept but stands out in its usage of both normative and pathological measures of identity. The study was exploratory yet grounded in well-validated theory. It also made use of valid and reliable questionnaires, administered to a large sample in the population of interest (i.e., emerging adults). The findings of the study could be both theoretically and clinically meaningful, but more research is needed to corroborate them.

Nonetheless, this study has several limitations. First, the data is cross-sectional. Therefore, it is not possible to draw conclusions about the directionality of associations. However, based on the

network model it is possible to make theoretically meaningful assumptions which can and should be tested using longitudinal and experimental research, which we suggest future research should do. Second, assessment of the variables was based on self-report measures. Third, the sample used in this study is diverse both in terms of age (i.e., late adolescents, emerging adults) and gender (an overrepresentation of women), which makes it more difficult to generalize the results to the entire population. With this sample size, we did not have the power to split the sample (i.e., between men and women; between adolescents and emerging adults), but future research should investigate age and gender differences in the identity-EMS relationship, for instance using networktree models (e.g., Hoorelbeke et al., 2023), as previous research has already demonstrated differences in identity, as previous research has already demonstrated differences in identity development can occur for these variables (Luyckx et al., 2013). The need for future research which tries to corroborate these preliminary findings while taking into account age and gender differences is extra important given that the study and its hypotheses were exploratory. Finally, the EMS entitlement had a low internal consistency (.57), which is not ideal, so conclusions about this schema should be made with caution.

## Conclusion

Using network-analysis, this study investigated the relationship between identity structures (identity synthesis, normative identity confusion, pathological identity confusion) and Young's EMS. The obtained network structure revealed unique associations between identity problems and EMS, mainly within the schema domain of disconnection/rejection and impaired limits. No unique associations between identity synthesis and EMS were found. These results indicate that EMS may be relevant factors in our understanding of maladaptive identity development, and vice versa. In particular, schemas relating to the theme of attachment and safe connection with others may be important, especially in our understanding of pathological identity problems.

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## Data availability statement

The data of this study will be made available upon reasonable request.

## References

- Ainsworth, M. S. (1979). Infant-mother attachment. *The American Psychologist*, 34(10), 932–937. <https://doi.org/10.1037//0003-066X.34.10.932>



- American Psychiatric Association. (2013). *Diagnostic and statistical manual of mental disorders* (5th ed.). Author. <https://doi.org/10.1176/appi.books.9780890425596>
- Arnett, J. J. (2000). Emerging adulthood: A theory of development from the late teens through the twenties. *The American Psychologist*, 55(5), 469–480. <https://doi.org/10.1037//0003-066X.55.5.469>
- Bach, B., Lockwood, G., & Young, J. E. (2017). A new look at the schema therapy model: Organization and role of early maladaptive schemas. *Cognitive Behaviour Therapy*, 47(4), 328–349. <https://doi.org/10.1080/16506073.2017.1410566>
- Bach, B., Simonsen, E., Christoffersen, P., & Kriston, L. (2015). The young schema questionnaire 3 short form (YSQ-S3). *European Journal of Psychological Assessment*. <https://doi.org/10.1027/1015-5759/a000272>
- Barazandeh, H., Kissane, D. W., Saeedi, N., & Gordon, M. (2016). A systematic review of the relationship between early maladaptive schemas and borderline personality disorder/traits. *Personality & Individual Differences*, 94, 130–139. <https://doi.org/10.1016/j.paid.2016.01.021>
- Beeney, J. E., Hallquist, M. N., Ellison, W. D., & Levy, K. N. (2016). Self-other disturbance in borderline personality disorder: Neural, self-report, and performance-based evidence. *Personality Disorders Theory, Research, & Treatment*, 7(1), 28. <https://doi.org/10.1037/per0000127>
- Berzonsky, M. D. (1990). Self-construction over the life-span: A process perspective on identity formation. In G. J. Neimeyer & R. A. Neimeyer (Eds.), *Advances in personal construct psychology: A research annual*, (Vol. 1, pp. 155–186). Elsevier Science/JAI Press.
- Beyers, W., & Goossens, L. (2008). Dynamics of perceived parenting and identity formation in late adolescence. *Journal of Adolescence*, 31(2), 165–184. <https://doi.org/10.1016/j.adolescence.2007.04.003>
- Beyers, W., Soenens, B., & Vansteenkiste, M. (2024). Autonomy in adolescence: A conceptual, developmental and cross-cultural perspective. *The European Journal of Developmental Psychology*, 22, 121–141. <https://doi.org/10.1080/17405629.2024.2330734>
- Bogaerts, A., Claes, L., Schwartz, S. J., Becht, A. I., Verschueren, M., Gandhi, A., & Luyckx, K. (2019). Identity structure and processes in adolescence: Examining the directionality of between- and within-person associations. *Journal of Youth & Adolescence*, 48(5), 891–907. <https://doi.org/10.1007/s10964-018-0931-5>
- Borsboom, D., Deserno, M. K., Rhemtulla, M., Epskamp, S., Fried, E. I., McNally, R. J., Dj, R., Perugini, M., Dalege, J., Costantini, G., Isvoranu Am Waldorp, L. J., Wysocki, A. C., van Borkulo, C. D., van Bork, R., & Waldorp, L. J. (2021). Network analysis of multivariate data in psychological science. *Nature Reviews Methods Primers*, 1(1), 58. <https://doi.org/10.1038/s43586-021-00055-w>
- Bosmans, G., Braet, C., & Van Vlierberghe, L. (2010). Attachment and symptoms of psychopathology: Early maladaptive schemas as a cognitive link? *Clinical Psychology & Psychotherapy*, 17(5), 374–385. <https://doi.org/10.1002/cpp.667>
- Bowlby, J. (1969). *Attachment and loss*. Random House.
- Chanen, A. M., & Thompson, K. N. (2019). The age of onset of personality disorders. *Age of Onset of Mental Disorders: Etiopathogenetic and Treatment Implications*, 183–201. [https://doi.org/10.1007/978-3-319-72619-9\\_10](https://doi.org/10.1007/978-3-319-72619-9_10)
- Crocetti, E., Klimstra, T., Keijsers, L., Hale, W. W., & Meeus, W. (2009). Anxiety trajectories and identity development in adolescence: A five-wave longitudinal study. *Journal of Youth & Adolescence*, 38, 839–849. <https://doi.org/10.1007/s10964-008-9302-y>
- Crocetti, E., Rubini, M., & Meeus, W. (2008). Capturing the dynamics of identity formation in various ethnic groups: Development and validation of a three-dimensional model. *Journal of Adolescence*, 31(2), 207–222. <https://doi.org/10.1016/j.adolescence.2007.09.002>
- Dozois, D. J., Martin, R. A., & Faulkner, B. (2013). Early maladaptive schemas, styles of humor and aggression. *Humor*, 26(1), 97–116. <https://doi.org/10.1515/humor-2013-0006>
- Epskamp, S., Borsboom, D., & Fried, E. I. (2018). Estimating psychological networks and their accuracy: A tutorial paper. *Behavior Research Methods*, 50, 195–212. <https://doi.org/10.3758/s13428-017-0862-1>
- Epskamp, S., Cramer, A. O., Waldorp, L. J., Schmittmann, V. D., & Borsboom, D. (2012). Qgraph: Network visualizations of relationships in psychometric data. *Journal of Statistical Software*, 48(4), 1–18. <https://doi.org/10.18637/jss.v048.i04>
- Epskamp, S., & Fried, E. I. (2018). A tutorial on regularized partial correlation networks. *Psychological Methods*, 23(4), 617. <https://doi.org/10.1037/met0000167>
- Epskamp, S., & Fried, E. I. (2020). Bootnet: Bootstrap methods for various network estimation routines. R package version 1.0. <https://CRAN.R-project.org/package=bootnet>
- Erikson, E. H. (1950). *Childhood and society*. W. W. Norton & Company.
- Esmaeilian, N., Dehghani, M., Koster, E. H., & Hoorelbeke, K. (2019). Early maladaptive schemas and borderline personality disorder features in a nonclinical sample: A network analysis. *Clinical Psychology & Psychotherapy*, 26(3), 388–398. <https://doi.org/10.1002/cpp.2360>
- Esmaeilian, N., Hoorelbeke, K., Naderzadeh, S., & Koster, E. H. (2022). Associations between borderline personality disorder features, early maladaptive schemas, and schema modes: A network analysis in a nonclinical sample. *Personality & Individual Differences*, 195, 111674. <https://doi.org/10.1016/j.paid.2022.111674>



- Friedman, E. J., Landsberg, A. S., Owen, J. P., Li, Y. O., & Mukherjee, P. (2014). Stochastic geometric network models for groups of functional and structural connectomes. *Neuroimage*, 101, 473–484. <https://doi.org/10.1016/j.neuroimage.2014.07.039>
- Fruchterman, T. M., & Reingold, E. M. (1991). Graph drawing by force-directed placement. *Software: Practice & Experience*, 21(11), 1129–1164. <https://doi.org/10.1002/spe.4380211102>
- Gandhi, A., Claes, L., Bosmans, G., Baetens, I., Wilderjans, T. F., Maitra, S., Kiekens, G., & Luyckx, K. (2016). Non-suicidal self-injury and adolescents attachment with peers and mother: The mediating role of identity synthesis and confusion. *Journal of Child & Family Studies*, 25, 1735–1745. <https://doi.org/10.1007/s10826-015-0350-0>
- Gandhi, A., Luyckx, K., Molenberghs, G., Baetens, I., Goossens, L., Maitra, S., & Claes, L. (2019). Maternal and peer attachment, identity formation, and non-suicidal self-injury: A longitudinal mediation study. *Child and Adolescent Psychiatry and Mental Health*, 13, 1–11. <https://doi.org/10.1186/s13034-019-0267-2>
- Hatano, K., Hihara, S., Nakama, R., Tsuzuki, M., Mizokami, S., & Sugimura, K. (2022). Trajectories in sense of identity and relationship with life satisfaction during adolescence and young adulthood. *Developmental Psychology*, 58(5), 977–989. <https://doi.org/10.1037/dev0001326>
- Hayward, L. E., Vartanian, L. R., Kwok, C., & Newby, J. M. (2020). How might childhood adversity predict adult psychological distress? Applying the identity disruption model to understanding depression and anxiety disorders. *Journal of Affective Disorders*, 265, 112–119. <https://doi.org/10.1016/j.jad.2020.01.036>
- Hoorelbeke, K., Faelens, L., De Raedt, R., & Koster, E. H. (2023). # ContextMatters! A network tree approach to model the link between social media use and well-being. *Computers in Human Behavior Reports*, 9, 100269. <https://doi.org/10.1016/j.chbr.2023.100269>
- Hoorelbeke, K., Marchetti, I., De Schryver, M., & Koster, E. H. (2016). The interplay between cognitive risk and resilience factors in remitted depression: A network analysis. *Journal of Affective Disorders*, 195, 96–104. <https://doi.org/10.1016/j.jad.2016.02.001>
- Hoorelbeke, K., Van den Bergh, N., Wichers, M., & Koster, E. H. (2019). Between vulnerability and resilience: A network analysis of fluctuations in cognitive risk and protective factors following remission from depression. *Behaviour Research and Therapy*, 116, 1–9. <https://doi.org/10.1016/j.brat.2019.01.007>
- Kaufman, E. A., Cundiff, J. M., & Crowell, S. E. (2015). The development, factor structure, and validation of the self-concept and identity measure (SCIM): A self-report assessment of clinical identity disturbance. *Journal of Psychopathology & Behavioral Assessment*, 37, 122–133. <https://doi.org/10.1007/s10862-014-9441-2>
- Kaufman, E. A., & Meddaoui, B. (2021). Identity pathology and borderline personality disorder: An empirical overview. *Current Opinion in Psychology*, 37, 82–88. <https://doi.org/10.1016/j.copsyc.2020.08.015>
- Keyfitz, L., Lumley, M. N., Hennig, K. H., & Dozois, D. J. A. (2013). The role of positive schemas in child psychopathology and resilience. *Cognitive Therapy and Research*, 37, 97–108. <https://doi.org/10.1007/s10608-012-9455-6>
- Lawrence, K. A., Allen, J. S., & Chanen, A. M. (2011). A study of maladaptive schemas and borderline personality disorder in young people. *Cognitive Therapy and Research*, 35, 30–39. <https://doi.org/10.1007/s10608-009-9292-4>
- Louis, J. P., Louis, K. M., & Young, A. M. (2023). Positive schemas: Replication, associations with negative schemas, and the dark triad. *Psychological Reports*, 126(6), 2856–2885. <https://doi.org/10.1177/00332941221100445>
- Louis, J. P., Wood, A. M., Lockwood, G., Ho, M. H. R., & Ferguson, E. (2018). Positive clinical psychology and schema therapy (ST): The development of the Young positive schema questionnaire (YPSQ) to complement the Young schema questionnaire 3 short form (YSQ-S3). *Psychological Assessment*, 30(9), 1199. <https://doi.org/10.1037/pas0000567>
- Luyckx, K., Goossens, L., Soenens, B., & Beyers, W. (2006). Unpacking commitment and exploration: Preliminary validation of an integrative model of late adolescent identity formation. *Journal of Adolescence*, 29(3), 361–378. <https://doi.org/10.1016/j.adolescence.2005.03.008>
- Luyckx, K., Klimstra, T. A., Duriez, B., Van Petegem, S., & Beyers, W. (2013). Personal identity processes from adolescence through the late 20s: Age trends, functionality, and depressive symptoms. *Social Development*, 22(4), 701–721. <https://doi.org/10.1111/sode.12027>
- Luyckx, K., Schwartz, S. J., Berzonsky, M. D., Soenens, B., Vansteenkiste, M., Smits, I., & Goossens, L. (2008). Capturing ruminative exploration: Extending the four-dimensional model of identity formation in late adolescence. *Journal of Research in Personality*, 42(1), 58–82. <https://doi.org/10.1016/j.jrp.2007.04.004>
- Maher, A., Cason, L., Huckstepp, T., Stallman, H., Kannis-Dymand, L., Millea, P., & Allen, A. (2022). Early maladaptive schemas in eating disorders: A systematic review. *European Eating Disorders Review*, 30(1), 3–22. <https://doi.org/10.1002/erv.2866>
- Marcia, J. E. (1966). Development and validation of ego-identity status. *Journal of Personality & Social Psychology*, 3(5), 551–558. <https://doi.org/10.1037/h0023281>
- Meeus, W. (2011). The study of adolescent identity formation 2000–2010. A review of longitudinal research. *Journal of Research on Adolescence*, 21(1), 75–94. <https://doi.org/10.1111/j.1532-7795.2010.00716.x>
- Nilsson, A. K. K., Jørgensen, C. R., Straarup, K. N., & Licht, R. W. (2010). Severity of affective temperament and maladaptive self-schemas differentiate borderline patients, bipolar patients, and controls. *Comprehensive Psychiatry*, 51(5), 486–491. <https://doi.org/10.1016/j.comppsy.2010.02.006>

- Rijkeboer, M. M., van den Bergh, H., & Van den Bout, J. (2005). Stability and discriminative power of the Young schema-questionnaire in a Dutch clinical versus non-clinical population. *Journal of Behavior Therapy and Experimental Psychiatry*, 36(2), 129–144. <https://doi.org/10.1016/j.jbtep.2004.08.005>
- Rosenthal, D. A., Moore, S. M., & Moore, S. M. (1981). From trust to intimacy: A new inventory for examining Erikson's stages of psychosocial development. *Journal of Youth & Adolescence*, 10(6), 525–537. <https://doi.org/10.1007/BF02087944>
- Schwartz, S. J. (2001). The evolution of Eriksonian and, neo-Eriksonian identity theory and research: A review and integration. *Identity*, 1(1), 7–58. <https://doi.org/10.1207/S1532706XSCHWARTZ>
- Schwartz, S. J., Beyers, W., Luyckx, K., Soenens, B., Zamboanga, B. L., Forthun, L. F., Sa, H., At, V., Ls, H., Sy, K., Sk, W., & Waterman, A. S. (2011). Examining the light and dark sides of emerging adults' identity: A study of identity status differences in positive and negative psychosocial functioning. *Journal of Youth & Adolescence*, 40, 839–859. <https://doi.org/10.1007/s10964-010-9606-6>
- Tariq, A., Quayle, E., Lawrie, S. M., Reid, C., & Chan, S. W. (2021a). Relationship between early maladaptive schemas and anxiety in adolescence and young adulthood: A systematic review and meta-analysis. *Journal of Affective Disorders*, 295, 1462–1473. <https://doi.org/10.1016/j.jad.2021.09.031>
- Tariq, A., Reid, C., & Chan, S. W. (2021b). A meta-analysis of the relationship between early maladaptive schemas and depression in adolescence and young adulthood. *Psychological Medicine*, 51(8), 1233–1248. <https://doi.org/10.1017/S0033291721001458>
- Thimm, J. C. (2009). Relationships between early maladaptive schemas and psychosocial developmental task resolution. *Clinical Psychology & Psychotherapy: An International Journal of Theory & Practice*, 17(3), 219–230. <https://doi.org/10.1002/cpp.647>
- Thimm, J. C. (2022). The higher-order structure of early maladaptive schemas: A meta-analytical approach. *Frontiers in Psychiatry*, 13, 1053927. <https://doi.org/10.3389/fpsy.2022.1053927>
- Vartanian, L. R., Hayward, L. E., Smyth, J. M., Paxton, S. J., & Touyz, S. W. (2018). Risk and resiliency factors related to body dissatisfaction and disordered eating: The identity disruption model. *International Journal of Eating Disorders*, 51(4), 322–330. <https://doi.org/10.1002/eat.22835>
- Verschuere, M., Claes, L., Bogaerts, A., Palmeroni, N., Gandhi, A., Moons, P., & Luyckx, K. (2018). Eating disorder symptomatology and identity formation in adolescence: A cross-lagged longitudinal approach. *Frontiers in Psychology*, 9, 816. <https://doi.org/10.3389/fpsyg.2018.00816>
- Verschuere, M., Luyckx, K., Kaufman, E. A., Vansteenkiste, M., Moons, P., Sleuwaegen, E., Berens, A., Schoevaerts, K., & Claes, L. (2017). Identity processes and statuses in patients with and without eating disorders. *European Eating Disorders Review*, 25(1), 26–35. <https://doi.org/10.1002/erv.2487>
- Videler, A. C., Hutsebaut, J., Schulkens, J. E., Sobczak, S., & Van Alphen, S. P. (2019). A life span perspective on borderline personality disorder. *Current Psychiatry Reports*, 21, 1–8. <https://doi.org/10.1007/s11920-019-1040-1>
- Westen, D., Betan, E., & DeFife, J. A. (2011). Identity disturbance in adolescence: Associations with borderline personality disorder. *Development & Psychopathology*, 23(1), 305–313.
- Young, J. E., & Arntz, A. (2014). *Schema-focused therapy for personality disorders*. Guilford Press.
- Young, J. E., & Brown, G. (2005). Young schema questionnaire-short form; version 3. *Psychological Assessment*. <https://doi.org/10.1037/t67023-000>
- Young, J. E., Klosko, J. S., & Weishaar, M. E. (2004). *Schema therapy: A practitioner's guide*. Guilford Press.
- Zeigler-Hill, V., Green, B. A., Arnau, R. C., Sisemore, T. B., & Myers, E. M. (2011). Trouble ahead, trouble behind: Narcissism and early maladaptive schemas. *Journal of Behavior Therapy and Experimental Psychiatry*, 42(1), 96–103. <https://doi.org/10.1016/j.jbtep.2010.07.004>
- Zhao, T., Liu, H., Roeder, K., Lafferty, J., Wasserman, L., & Zhao, M. T. (2015). Hugu: High-dimensional undirected graph estimation. R package version 1.3.5. <https://CRAN.R-project.org/package=hugu>